

A Full Answer to the ℓ_1 -Copy Question in arXiv:2605.21279

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Source

Verónica Dimant, Luis C. García-Lirola, Juan Guerrero-Viu, and Antonín Procházka, *Composition operators for holomorphic Lipschitz functions*, arXiv:2605.21279.

The question appears in Remark 4.21 on page 25 of the source PDF.

Since the weak topology of $\mathcal{A}_{wu}(B_X, X^*)$ is induced by the one of $C((\overline{B}_X, w), X^*)$, we equivalently have to prove the weak convergence in this latter space. By the compactness of $(\overline{B}_X, w)$ and the reflexivity of X we have that the set $\{\delta(x) \otimes y : x, y \in \overline{B}_X\}$ is a James boundary for $C((\overline{B}_X, w), X^*)$. Hence, by Rainwater-Simons theorem [27, Theorem 3.134] it is enough to see that $C_\phi^{X^*}(f_{n_k})(x)(y) \xrightarrow{w} C_\phi^{X^*}(g)(x)(y)$, for all $x, y \in \overline{B}_X$. This is clear from the previous arguments since $C_\phi^{X^*}(f_{n_k})(x)(y) = f_{n_k}(x/2)(y)$ and $C_\phi^{X^*}(g)(x)(y) = g(x/2)(y)$.

Remark 4.21. Recall that, by Lemma 3.2 we have $C_\phi^L = \widehat{\phi}^*$. Hence, by the Schauder theorem, the Gantmacher theorem and Lemma 2.5, the results in this section concerning compactness, weak compactness, and ℓ_∞ -strict singularity of C_ϕ^L can equivalently be formulated in terms of compactness, weak-compactness, and the property of fixing no complemented copies of ℓ_1 for $\widehat{\phi}$, respectively.

We do not know, however, the answer to the following question: if $\widehat{\phi}$ fixes a copy of ℓ_1 , must it fix a complemented copy of ℓ_1 ? Note that this implication holds for linearizations of Lipschitz maps, see [37].

Problem Statement

For a holomorphic Lipschitz symbol ϕ , the paper writes

$$C_\phi^L = \widehat{\phi}^*,$$

where

$$\widehat{\phi} : \mathcal{G}_0(B_Y) \longrightarrow \mathcal{G}_0(B_X)$$

is the corresponding holomorphic-Lipschitz linearization. Remark 4.21 asks: if $\widehat{\phi}$ fixes a copy of ℓ_1 , must it fix a complemented copy of ℓ_1 ?

Result

Theorem 1. *Let $T : Y \rightarrow X$ be a bounded linear operator between Banach spaces. If T fixes a copy of ℓ_1 , then T fixes a complemented copy of ℓ_1 .*

Proof. Assume that T fixes a copy of ℓ_1 . Then there is an isomorphic embedding

$$A : \ell_1 \longrightarrow Y$$

such that $TA : \ell_1 \rightarrow X$ is also an isomorphic embedding.

Consider the adjoint

$$Q = (TA)^* : X^* \longrightarrow (\ell_1)^* = \ell_\infty.$$

This map is onto. Indeed, a functional on ℓ_1 defines a bounded functional on the subspace $(TA)(\ell_1) \subseteq X$ via $(TA)^{-1}$, and Hahn–Banach extends it to all of X .

Since ℓ_∞ is injective, the quotient map Q has a bounded linear right inverse. Thus there is a bounded operator

$$J : \ell_\infty \longrightarrow X^*$$

such that $QJ = I_{\ell_\infty}$.

Restrict J to the canonical copy of $c_0 \subseteq \ell_\infty$. For every $u \in c_0$,

$$u = QJu = (TA)^*Ju = A^*T^*Ju.$$

It follows that $T^*J|_{c_0} : c_0 \rightarrow Y^*$ is an isomorphic embedding: the upper estimate is immediate from boundedness, while the lower estimate follows from

$$\|u\| \leq \|A^*\| \|T^*Ju\|.$$

Therefore $T^* : X^* \rightarrow Y^*$ fixes a copy of c_0 .

Lemma 2.5 of the source paper states that, if T^* fixes a copy of c_0 , then T fixes a complemented copy of ℓ_1 . Applying that lemma gives the desired complemented copy. $\square$

Corollary 1. *Let X, Y be complex Banach spaces and let $\phi \in \mathcal{HL}_0(B_Y, X)$ satisfy $\phi(B_Y) \subseteq B_X$. If the holomorphic-Lipschitz linearization*

$$\widehat{\phi} : \mathcal{G}_0(B_Y) \rightarrow \mathcal{G}_0(B_X)$$

fixes a copy of ℓ_1 , then $\widehat{\phi}$ fixes a complemented copy of ℓ_1 .

Proof. Apply the theorem to $T = \widehat{\phi}$. $\square$

Verification Notes

The proof uses three standard ingredients.

- If $TA : \ell_1 \rightarrow X$ is an isomorphic embedding, then $(TA)^* : X^* \rightarrow \ell_\infty$ is onto by Hahn–Banach.
- The space ℓ_∞ is injective, so this quotient map splits.
- Lemma 2.5 of arXiv:2605.21279 says that T^* fixing c_0 implies T fixes a complemented copy of ℓ_1 .

No computational verification is relevant.

Limitations and Novelty Check

The argument is a general operator-theoretic observation and does not use any special geometry of holomorphic-Lipschitz free spaces. For that reason, the result should be reviewed as a short deduction from classical facts together with Lemma 2.5 of the source paper.

Verdict

The question in Remark 4.21 has an affirmative answer as stated: $\widehat{\phi}$ fixing a copy of ℓ_1 automatically implies that $\widehat{\phi}$ fixes a complemented copy of ℓ_1 .